

Programming Using C++ Level 1

Conditional statements



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Definition

- ◆ A conditional statement allows a program to make decisions based on whether a condition is true or false.
- ◆ In C++, if and else are commonly used to execute different code depending on a condition.

Explanation

- ◆ This program checks two conditions:
 - The age must be greater than or equal to 21.
 - The user must have a driver license and enter "yes".
- ◆ The logical AND operator && requires both conditions to be true.
- ◆ If both conditions are true, the program prints Hired.
- ◆ Otherwise, it prints Rejected.

Steps in main()

1. Get the user's age

- ◆ Store the entered age in the age variable.

2. Get the driver's license information

- ◆ Store the user's answer in hasDriverLicense.

3. Check the conditions

- ◆ Check whether `age >= 21` and `hasDriverLicense == "yes"`.

4. Execute the if statement

- ◆ If both conditions are true, print `Hired`.

5. Execute the else statement

- ◆ If the condition is false, print `Rejected`.

💡 Example

◆ Condition:

if (age >= 21 && hasDriverLicense == "yes")

◆ If:

age = 25

hasDriverLicense = "yes"

◆ Both conditions are true, so the result is:

Hired

➡ Result

◆ If the user enters:

Age: 25

Driver license: yes

◆ The program displays:

Hired

◆ If the user enters:

Age: 20

Driver license: yes

◆ The program displays:

Rejected

Note

- ◆ `>=` means greater than or equal to.
- ◆ `==` compares two values.
- ◆ `&&` means AND, so both conditions must be true.
- ◆ The function `checkEligibility()` is responsible for making the decision, while `getUserInput()` is responsible for getting the user's input.

⊘ Warning

- ◆ Do not use `=` when you want to compare values.
- ◆ Use `==` for comparison and `=` for assignment.



Comparison



if

- ✓ Executes code when a condition is true.
- ✓ Used to make decisions.
- ✓ Can be combined with logical operators.



else

- ✓ Executes when the if condition is false.
- ✓ Provides an alternative result.
- ✓ Does not have its own condition.

✦✦ Summary

- ✓ Conditional statements allow programs to make decisions.
- ✓ if executes code when a condition is true, while else handles the false case.
- ✓ Logical operators such as && can combine multiple conditions.

Thank you 😊